```
# ============================================================
# CONCEPT MAP: IMAGE SENSING MECHANISMS
# Google Colab Code
# ============================================================

# Install Graphviz
!apt-get -qq install graphviz
!pip -q install graphviz

from graphviz import Digraph
from IPython.display import Image, display

# ------------------------------------------------------------
# Create Concept Map
# ------------------------------------------------------------
dot = Digraph("ImageSensingMechanisms", format="png")

# Layout Settings
dot.attr(rankdir="TB")
dot.attr(size="16,16")
dot.attr(dpi="300")
dot.attr(nodesep="0.5")
dot.attr(ranksep="0.8")
dot.attr(splines="ortho")

# ============================================================
# MAIN NODE
# ============================================================
```

```
dot.node(
    "A",
    "IMAGE SENSING\nMECHANISMS",
    shape="box",
    style="filled",
    fillcolor="lightblue",
    fontsize="22"
)

# ============================================================
# IMAGE SENSING COMPONENTS
# ============================================================

dot.node("B",
         "Optical Lens",
         shape="box",
         style="filled",
         fillcolor="lightgreen")

dot.node("C",
         "CCD / CMOS Sensor",
         shape="box",
         style="filled",
         fillcolor="lightgreen")

dot.node("D",
         "Pixel Array",
         shape="box",
         style="filled",
         fillcolor="lightgreen")
```

```
dot.node("E",
         "Analog-to-Digital\nConverter (ADC)",
         shape="box",
         style="filled",
         fillcolor="lightgreen")

# ============================================================
# SENSOR CHARACTERISTICS
# ============================================================

dot.node("F",
         "Sensor Characteristics",
         shape="box",
         style="filled",
         fillcolor="khaki")

dot.node("G","Pixel Size",shape="ellipse")
dot.node("H","Quantum Efficiency",shape="ellipse")
dot.node("I","Dynamic Range",shape="ellipse")
dot.node("J","Sensor Sensitivity (ISO)",shape="ellipse")

# ============================================================
# NOISE SOURCES
# ============================================================

dot.node("K",
         "Noise Generation",
         shape="box",
         style="filled",
         fillcolor="lightcoral")
```

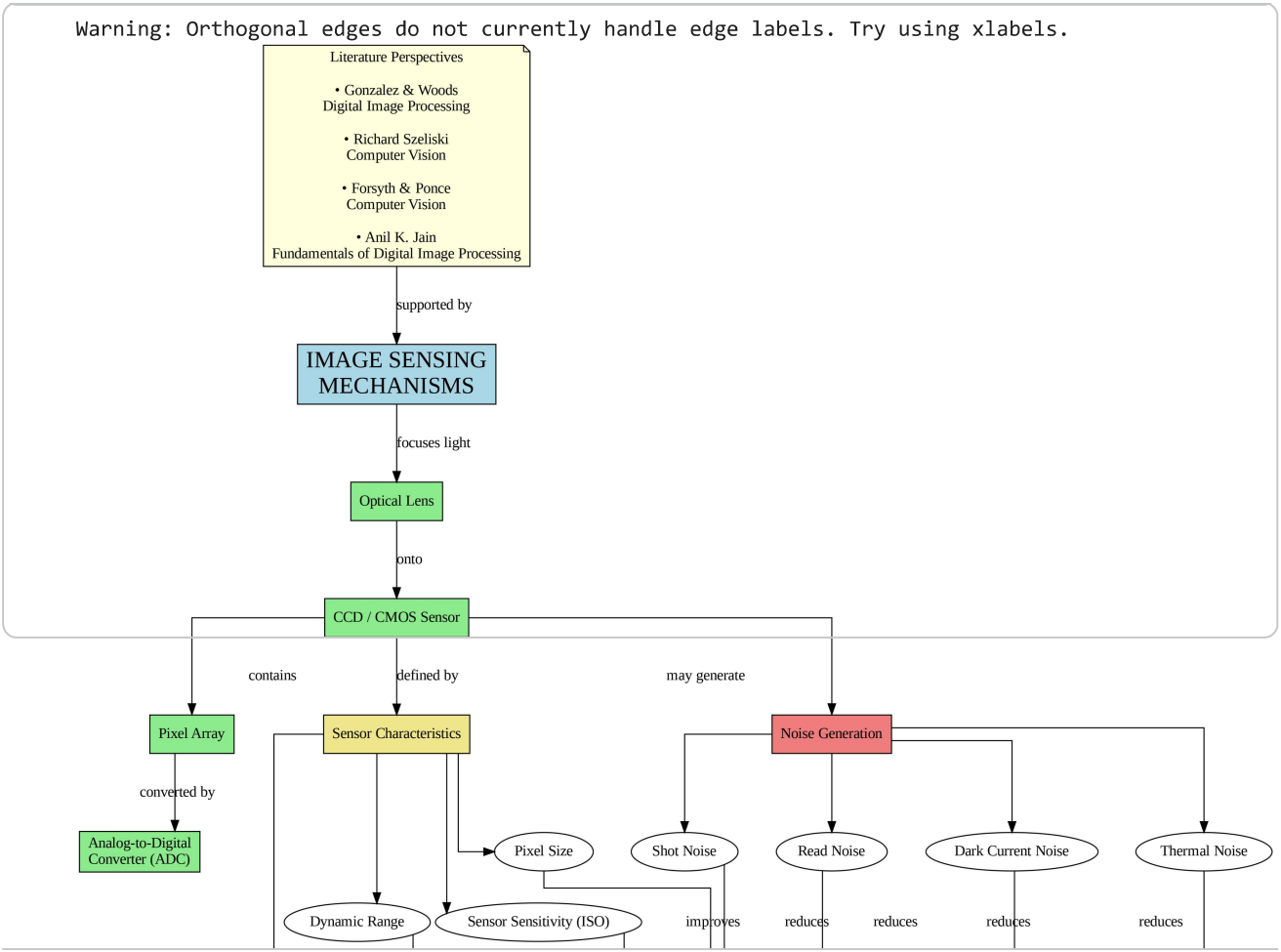
```
dot.node("L","Thermal Noise",shape="ellipse")
dot.node("M","Shot Noise",shape="ellipse")
dot.node("N","Read Noise",shape="ellipse")
dot.node("O","Dark Current Noise",shape="ellipse")

# ============================================================
# IMAGE ACQUISITION QUALITY
# ============================================================

dot.node("P",
         "Image Acquisition Quality",
         shape="box",
         style="filled",
         fillcolor="lightskyblue")

dot.node("Q","High Signal-to-Noise Ratio",shape="ellipse")
dot.node("R","High Resolution",shape="ellipse")
dot.node("S","Accurate Color Reproduction",shape="ellipse")
dot.node("T","Improved Image Quality",shape="ellipse")

# ============================================================
# APPLICATIONS
# ============================================================

dot.node("U",
         "Computer Vision Applications",
         shape="box",
         style="filled",
         fillcolor="palegreen")

dot.node("V","Medical Imaging",shape="box")
dot.node("W","Autonomous Vehicles",shape="box")
```

```
dot.node("W","Autonomous Vehicles",shape="box")
dot.node("X","Remote Sensing",shape="box")
dot.node("Y","Face Recognition",shape="box")
dot.node("Z","Industrial Inspection",shape="box")

# ============================================================
# LITERATURE
# ============================================================

dot.node(
    "AA",
    "Literature Perspectives\n\n"
    "• Gonzalez & Woods\nDigital Image Processing\n\n"
    "• Richard Szeliski\nComputer Vision\n\n"
    "• Forsyth & Ponce\nComputer Vision\n\n"
    "• Anil K. Jain\nFundamentals of Digital Image Processing",
    shape="note",
    style="filled",
    fillcolor="lightyellow"
)

# ============================================================
# CONNECTIONS
# ============================================================

# Main workflow
dot.edge("A","B",label="focuses light")
dot.edge("B","C",label="onto")
dot.edge("C","D",label="contains")
dot.edge("D","E",label="converted by")

# Sensor characteristics
```

```
dot.edge("C","F",label="defined by")
dot.edge("F","G")
dot.edge("F","H")
dot.edge("F","I")
dot.edge("F","J")

# Characteristics → Quality
dot.edge("G","P",label="improves")
dot.edge("H","Q",label="increases")
dot.edge("I","R",label="supports")
dot.edge("J","S",label="affects")

# Noise
dot.edge("C","K",label="may generate")
dot.edge("K","L")
dot.edge("K","M")
dot.edge("K","N")
dot.edge("K","O")

# Noise impact
dot.edge("L","P",label="reduces")
dot.edge("M","P",label="reduces")
dot.edge("N","P",label="reduces")
dot.edge("O","P",label="reduces")

# Acquisition Quality
dot.edge("P","Q")
dot.edge("P","R")
dot.edge("P","S")
dot.edge("Q","T")
dot.edge("R","T")
```

```
dot.edge("S","T")

# Applications
dot.edge("T","U",label="supports")
dot.edge("U","V")
dot.edge("U","W")
dot.edge("U","X")
dot.edge("U","Y")
dot.edge("U","Z")

# Literature
dot.edge("AA","A",label="supported by")

# ============================================================
# SAVE & DISPLAY
# ============================================================

filename = dot.render("Image_Sensing_Mechanisms_Concept_Map")

display(Image(filename))

print("Concept Map generated successfully!")
```

```
Warning: Orthogonal edges do not currently handle edge labels. Try using xlabels.
```



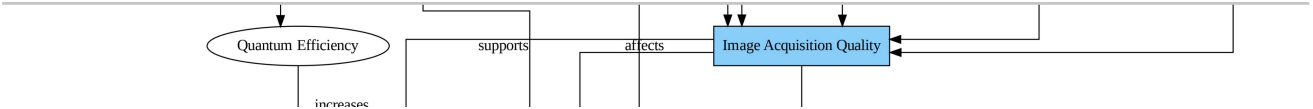

Quantum Efficiency
supports
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Image Acquisition Quality
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